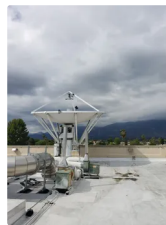




Proposal (Updated January 2023)

Restoring the Moore Lab Radio Telescope

Michael Gutierrez, Saren Daghljan
(Updated January 2023)



Elevator Pitch:

- This 6-meter radio telescope on the roof of Moore Building was donated to Caltech as a prototyping and educational tool for radio astronomy.
- It has not been used for ~7 years, and its previous administrator Sander Weinreb has no further use for it.
- An operational radio telescope managed and maintained by students would enable an endless variety of learning and research opportunities for the Caltech community.

History of the telescope:

2002-2004: Initial concept and design of the antenna for the purpose of extending the Deep Space Network array. A few prototypes were built and used for a Caltech-JPL interferometry experiment[1][2][3]

2005: One of these telescopes was donated to Caltech by Charles Elachi and installed on the roof of Moore Lab

2013: Subreflector removed and new QRFH prime focus feed designed by Akgiray and Weinreb[4]

2014-2015: Hardware and software set up and calibrated for basic astronomy demonstrations under Weinreb[5]

2016-Present: Astronomy department moves out of Moore Building; most of the telescope's existing infrastructure was lost.

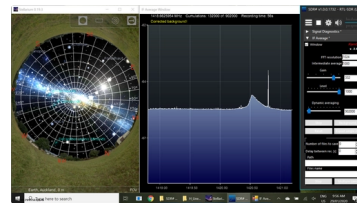
The Idea:

We want to turn the Moore Radio Telescope into a tool for astronomy and radio science education. It would be available to students and faculty at Caltech to use for teaching, research, club/personal projects, and outreach.

We have spoken to several undergraduates who would be interested in heading up this project during their free time. Additionally, Profs. Charles Elachi, Bethany Ehlmann, and Sander Weinreb, as well as Andrew Klesh and Lisa Locke at JPL, have expressed interest in supporting or collaborating with a student-run radio telescope project.

This project will be feasible within a very reasonable timeline and budget. This is due in part to software defined radio (SDR) technology, which has exploded in popularity in the past few years, leading to a wealth of open-source software (and hardware). For example, MIT Haystack Observatory's Small Radio Telescope project^[8] utilized an RTL-SDR (below) and was duplicated by students at hundreds of universities around the world.

Cheap (<\$30) commercial off-the-shelf electronics make radio astronomy accessible to amateurs^{[6][7]}



With the proposed upgrades (see The Game Plan below), the telescope will be remotely operable from a web interface. Interested parties can schedule observation time after being trained in proper operation of the telescope.

Immediate use cases:

- EE/Ay project classes: the telescope's electronics were built in a configurable "breadboard" layout specifically for testing new radio frequency (RF) components.
- Radio Astronomy: follow-up observations on possible sources of fast radio bursts (FRBs) in conjunction with Owens Valley Radio Observatory (OVRO).
- Satellite Communications: downlink data from satellites as they pass overhead. Provides learning opportunities for RF communications / digital signal processing (DSP) as well as spacecraft mission operations. Possible collaboration with JPL, Caltech CubeSat Club, Amateur Radio Club (see The Future below).

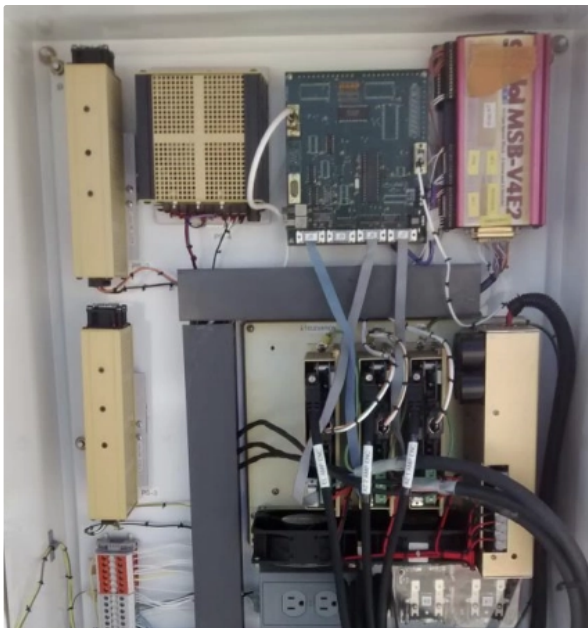
The Game Plan:

We have received permission to access the roof and the telescope by the EE Department Administrator, Mabby Howard. Prof. Weinreb has been advising our work. Most of the vital components (reflector, servos, control electronics) are still in good shape, and we have been able to connect to the telescope computer and move the dish. The parts that were lost/damaged (RF amplifiers, spectrum analyzer, filters) will be relatively easy and inexpensive to replace.

To do:

- Replace the USB-to-serial adapter for the servo controller and test rotators
- We need to reposition the antenna to access the feed, so a good first test will be to establish a serial connection with the controller and try moving it.
- One of the redundant azimuth motors is gone, the whole thing needs a lot of lubrication, and the brakes will need to be disabled first, but otherwise it should be in working order.
- Closer inspection of the mechanicals + the feed
- Will require setup of harness safety system or a large stepladder
- Rewire the RF pathway (likely replace the front-end amplifier)
- Many old parts are not salvageable - new components and cabling needed
- Create a fork of MIT's SRT-Py code^[9] (a convenient user interface for radio telescopes) to add compatibility for our model of servo controller
- We found a laptop with the original rotator control code which we plan to reverse-engineer.
- Calibrate motor encoders and spectrum analyzer
- In order to accurately point at and measure sky objects. Mostly trial-and-error.

Inside the control box of the telescope



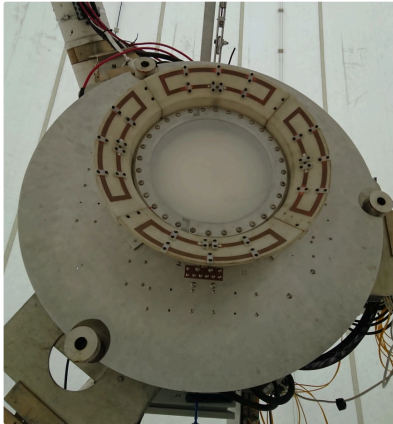
Current Roadblocks:

- In need of lab equipment (oscilloscope, network analyzer, etc) to continue work on the RF components of the telescope. So far, we have been borrowing this equipment, which is not sustainable.
- Funding for new components: planning to apply for a Housner Grant through the Deans' Office as well as an [ARDC Grant](#)
- Would be nice to have a faculty/staff advisor in the EE dept. to supervise safe physical work on the telescope and interface with instead of bugging the department administrator.

The Future:

We could greatly expand the functionality of this telescope by purchasing or building a new feed with both receive and transmit capabilities. This would likely require a significant amount of time and funding, but it would make for an excellent student research project. A similar project was done by a graduate student at MIT Haystack Observatory, albeit for a much larger-scale application (see [10]).

A novel radio feed designed for satellite communications at MIT's Westford Radio Telescope



Why go through the trouble to add a transmitter?

- Satellite communications: With TX and RX capability, this telescope could function as a highly effective ground station for spacecraft command and control. There are several organizations at Caltech that launch satellites on a semi-regular basis, and a private on-campus ground station would be extremely convenient for them. (In particular, our undergraduate CubeSat Club and JPL's SmallSats Student Mission Operations Center.)
- Additionally, it would be interesting to collaborate with the Naval Postgraduate School's Mobile CubeSat Command & Control project.[11] This is a global network of satellite ground stations that supports academic and scientific spacecraft.
- Amateur radio applications: Caltech's Amateur Radio Club is also potentially interested in working on this telescope. Potential uses include long-distance communications via ionospheric reflections and possibly even moon-bounce experiments.[12]

Further Reading:

[1] [The DSN Array Development Program](#)

[2] [The 6-meter Breadboard Antenna for the Deep Space Network Large Array](#)

[3] [An Antenna Servo Test Bed for the Deep Space Array Network](#)

[4] [Akgiray PhD Thesis](#)

[5] [Caltech 6m Antenna Hydrogen Line Report](#)

[6] [Cheap and Easy Hydrogen Line Radio Astronomy with an RTL-SDR](#)

[7] [A low-cost 21 cm horn-antenna radio telescope for education and outreach](#)

[8] [SRT: The Small Radio Telescope - MIT Haystack Observatory](#)

[9] [MITHaystack/srt-py](#)

[10] [A UHF Multimode Array Feed for the Westford Radio Telescope](#)

[11] [The Mobile CubeSat Command and Control \(MC3\) Ground Station Network](#)

[12] [Moonbounce \(EME\) - Radio Society of Great Britain - Main Site](#)